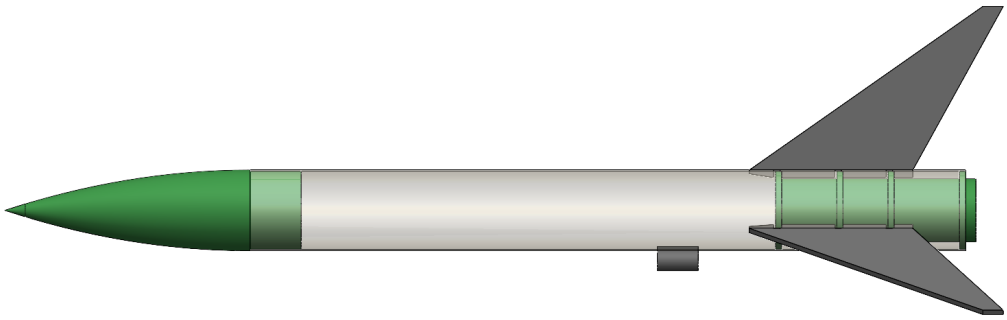
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Engineering Portfolio

Texas A&M, Aerospace Engineering



2026

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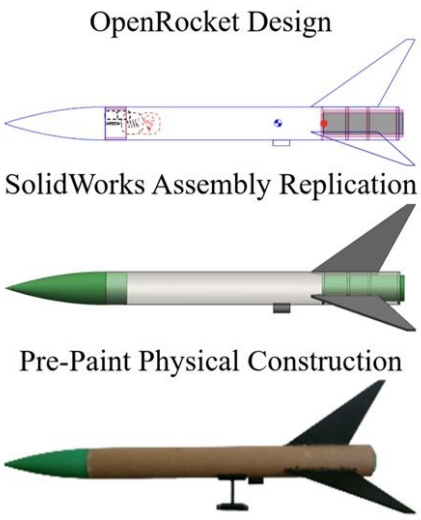
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E-Class Model Rocket Design & Fabrication

May 2026 Tools: SolidWorks, OpenRocket, Bambu Lab P1S



Development Process



Flight Test Launch

Predicted Apogee	Actual Apogee	Error
1,227 ft	1,064 ft	15.3%

Objective

Engineer a custom E-class model rocket by executing a complete design life cycle using aerodynamic simulation, CAD modeling, additive manufacturing, assembly, and empirical flight test validation.

Engineering Highlights

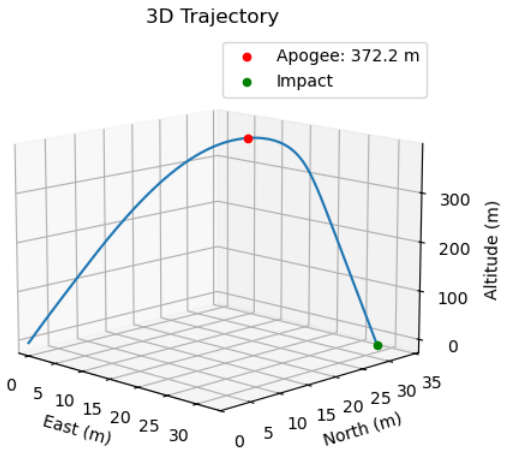
- **Flight Dynamics & Trajectory Simulation:** Conducted rigorous flight stability and trajectory simulations using OpenRocket to optimize fin geometry, body tube length, nose cone shape, and overall weight distribution. These simulations allowed validation of the design stability and predicted apogee prior to any manufacturing or assembly. The simulations also validated critical launch aspects such as off-rail velocity, timing of chute deployment, and descent speed to ensure safe operations.
- **Parametric CAD & Additive Manufacturing:** Replicated OpenRocket design as a 3D-assembly in SolidWorks. Created a through-the-wall fin attachment mechanism with friction fitting precision to facilitate easy fin alignment during fabrication. In addition, a custom motor mount was created to hold the fins in alignment and keep the motor in a stable position during flight. All parts were designed for rapid additive manufacturing on a Bambu Lab P1S with peak loads during flight in mind.
- **System Assembly:** Assembled 3D-printed parts and cardboard body tube as designed in previous phases of the design cycle. Securely routed and anchored the shock cord to withstand peak deployment shock loads, and methodically packed the parachute with protective wadding to ensure a reliable, damage-free deployment from the motor's ejection charge.

Results

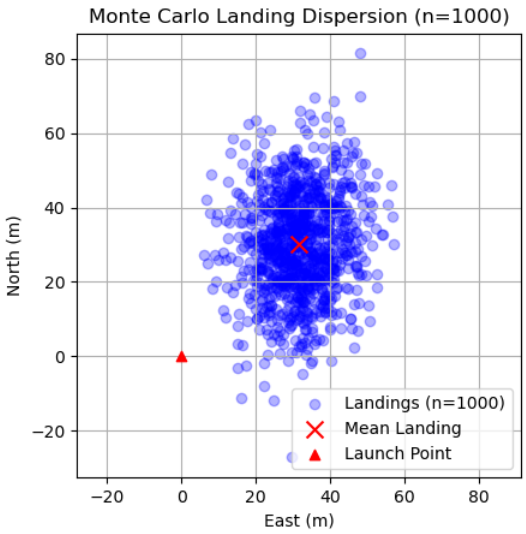
Achieved a successful test flight with nominal parachute deployment and recovery. An on board altimeter measured an apogee (1,064 ft) which closely reflected the pre-flight simulation apogee (1,227 ft). The 15.3% margin of error was likely due to weather cocking, unmodeled parasitic drag, and variations in the actual total impulse of the commercial motor.

3-DOF Rocket Flight Simulator & Monte Carlo Dispersion Analysis

August 2026 Tools: Python, NumPy, Matplotlib, PyYAML, pytest



3D Flight Trajectory



Monte Carlo Landing Dispersion ($n = 1{,}000$)

OpenRocket Apogee	Simulated Apogee	Error
376 m	372.2 m	1.0%

Objective

Build a first-principles 3-DOF flight simulator, driven by real motor and atmospheric data, capable of independently reproducing a validated rocket's ascent performance and characterizing its landing footprint through statistical dispersion analysis – the same class of methodology used in real range-safety studies.

Engineering Highlights

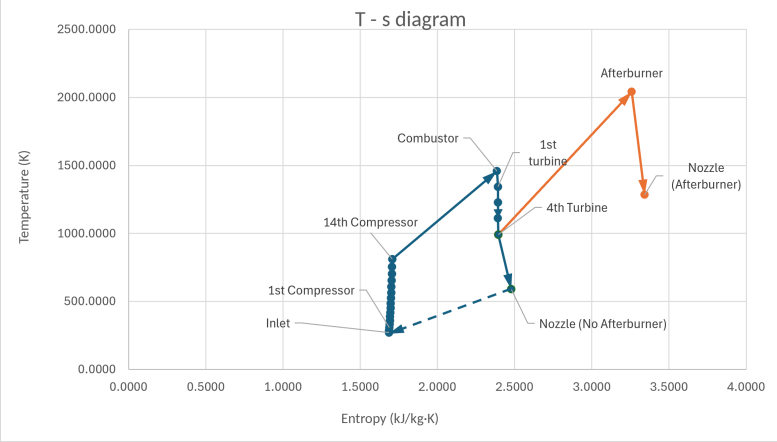
- **Physics Modeling & Software Architecture:** Modeled point-mass dynamics in East/Altitude/North space under gravity, thrust, drag, and wind, with thrust and wind independently resolved into magnitude and azimuth components, using real motor thrust curves loaded from `.eng` files. Built a full 3-layer International Standard Atmosphere model from first principles rather than a lookup table, and validated against reference density values. Refactored the simulation into a single callable function and decoupled every rocket's parameters into per-rocket YAML configuration files, requiring no code changes for new designs.
- **Recovery System Modeling:** Implemented three configurable recovery modes – ballistic, single-deployment, and dual-deployment – with apogee and altitude-triggered deployment events detected in real time during integration via velocity sign change and altitude threshold, rather than inferred after the fact.
- **Monte Carlo Dispersion Analysis:** Batch-simulated 1,000 trials per rocket, independently randomizing wind speed, wind direction, launch angles, and drag coefficient around each rocket's nominal configuration. Derived wind-speed uncertainty directly from each launch site's reported sustained-vs-gust wind spread rather than an arbitrary fixed value, so gustier conditions produce proportionally wider dispersion.
- **Validation:** Cross-validated ascent performance against an independent OpenRocket simulation of the same rocket and motor, and confirmed thrust and wind decomposition against hand-calculated expectations.

Results

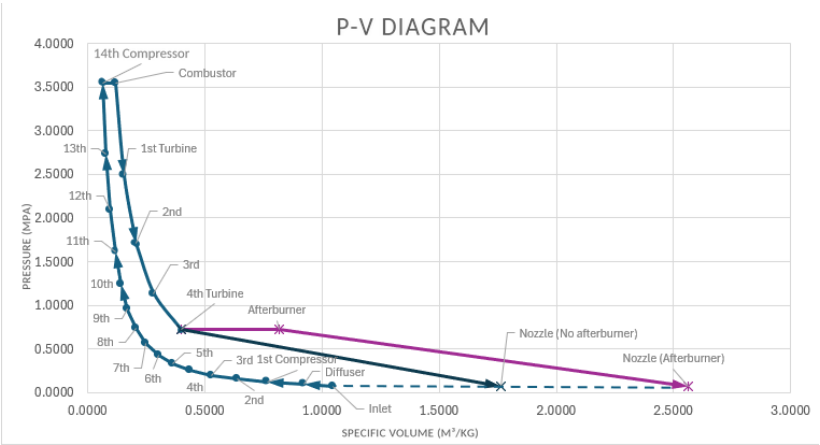
The simulator reproduced OpenRocket's predicted apogee to about a 1.0% error on Plus Ultra 1.0, a real custom E-class rocket flown on an Estes E12 motor. A 1,000-run Monte Carlo dispersion analysis on the same rocket characterized both apogee variance and landing footprint, demonstrating a complete range-safety-style dispersion workflow end-to-end.

Turbojet Engine Thermodynamic Analysis

December 2025 Tools: Excel, Technical Reporting



T-s Diagram Analysis



P-v Diagram Analysis

Engine Mode	Thrust (kN)
No Afterburner	61.394
With Afterburner	95.271

Objective

Analyze the thermodynamic performance of an ideal Brayton-cycle turbojet engine.

Engineering Highlights

- Modeled the operational efficiency of the compressor, combustor, turbine, and nozzle stages.
- Plotted comprehensive T-s and P-v diagrams in Excel to visualize cycle trends.
- Evaluated a secondary afterburner configuration to quantify deviations in thrust output.
- Benchmarked analytical results against published Pratt and Whitney F-100 reference data.

Results

Predicted engine thrust within approximately 10% of reference performance data, successfully validating the air-standard assumptions.